EXCLUSIVE FEATURE

The Open-Source Engine That Could Save $47B in Data Center Power

How a 613,000-parameter neural network is rewriting the rules of GPU data center energy management - and why ERCOT just made it essential.

30%	59%	<21s	100%
GRID SMOOTHING	VARIANCE REDUCTION	RESPONSE TIME	OPEN SOURCE

Inside This Issue

A deep dive into the technology, the market, and the mission.

About This Publication

This is a special feature showcasing Energivanu, an open-source ML toolkit for GPU data center power optimization. Built on real data, validated on real hardware, designed for the ERCOT PCLR era.

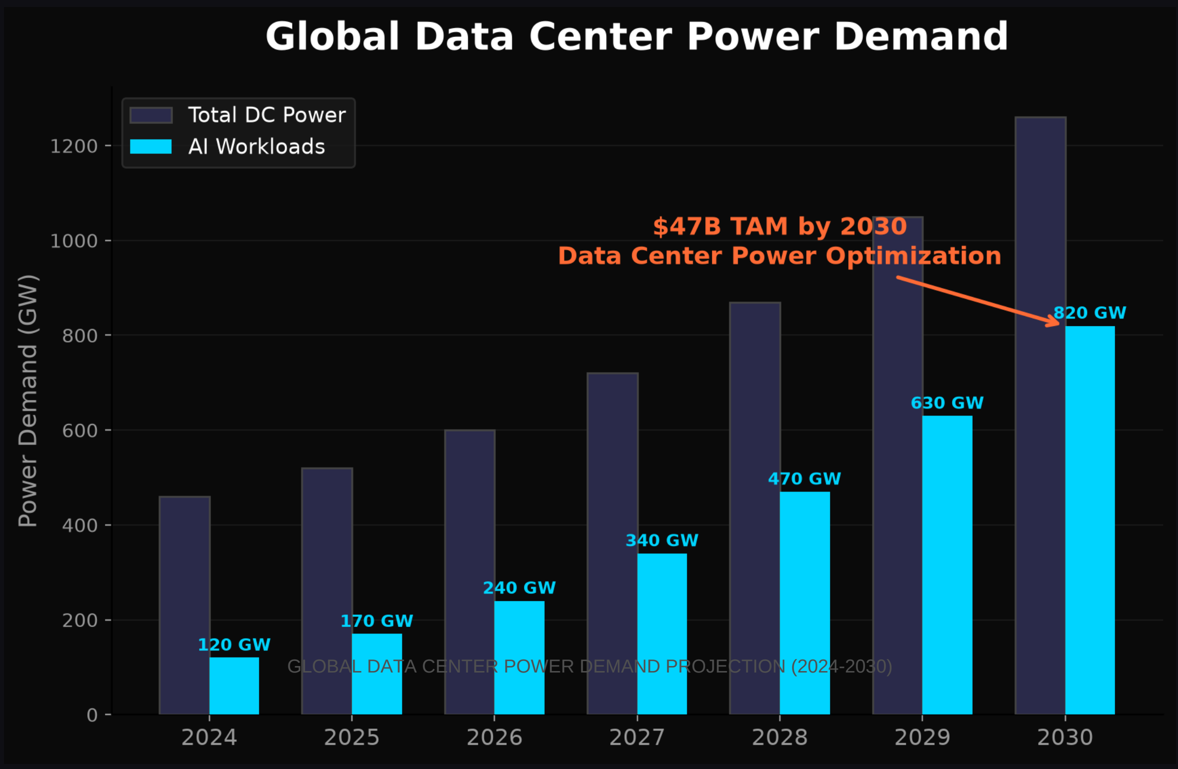
The $47 Billion Problem

AI's insatiable appetite for power is creating a crisis that grid operators can no longer ignore.

> When ERCOT approved the Passive Controllable Load Resource framework on June 18, 2026, it wasn't just creating a new regulatory category - it was acknowledging a fundamental shift in how the electrical grid interacts with the largest power consumers on the planet.

The numbers are staggering. Global data center power demand is projected to reach 1,260 GW by 2030, with AI workloads consuming an estimated 820 GW of that total. The ERCOT grid alone faces a 410 GW large-load queue - and 87% of those applications are for data centers. These aren't incremental additions to the grid. They're entire power plants worth of demand, arriving faster than utilities can build transmission lines.

The problem isn't just raw power consumption. It's volatility. When a distributed AI training job triggers an All-Reduce synchronization across thousands of GPUs, the power draw can spike by megawatts in seconds. These spikes stress transformers, trip breakers, and force utilities to maintain expensive spinning reserves just to handle the turbulence.



Enter Energivanu

The open-source execution engine that fills the gap between grid signals and GPU clusters.

> Energivanu is what happens when you treat data center power optimization as a machine learning problem instead of a facilities management problem. The result is a 613,000-parameter neural network that can predict power spikes, dispatch battery storage, and stagger training phases - all in under 21 seconds.

The project started with a simple observation: GPU data centers don't have a power problem. They have a coordination problem. When a thousand GPUs start an All-Reduce operation simultaneously, the power spike isn't just large - it's unpredictable. Traditional demand response systems, designed for HVAC loads and lighting schedules, can't handle millisecond-scale volatility from compute workloads.

Energivanu approaches this differently. Instead of treating the data center as a black box with a power meter, it reads the training telemetry directly - GPU utilization, temperature, memory bandwidth, communication patterns - and uses that signal to predict what the power draw will look like 10 steps into the future.

613K PARAMETERS TCN + Attention	**15** INPUT FEATURES Power - GPU - System	**10** PREDICTION STEPS Future Power Forecast
21% MAPE Alibaba 30L Rows	**<21s** RESPONSE 30x Faster Than PCLR	**AGPL** LICENSE Open Source

Three Engines, One Pipeline

1. Power Prediction Engine

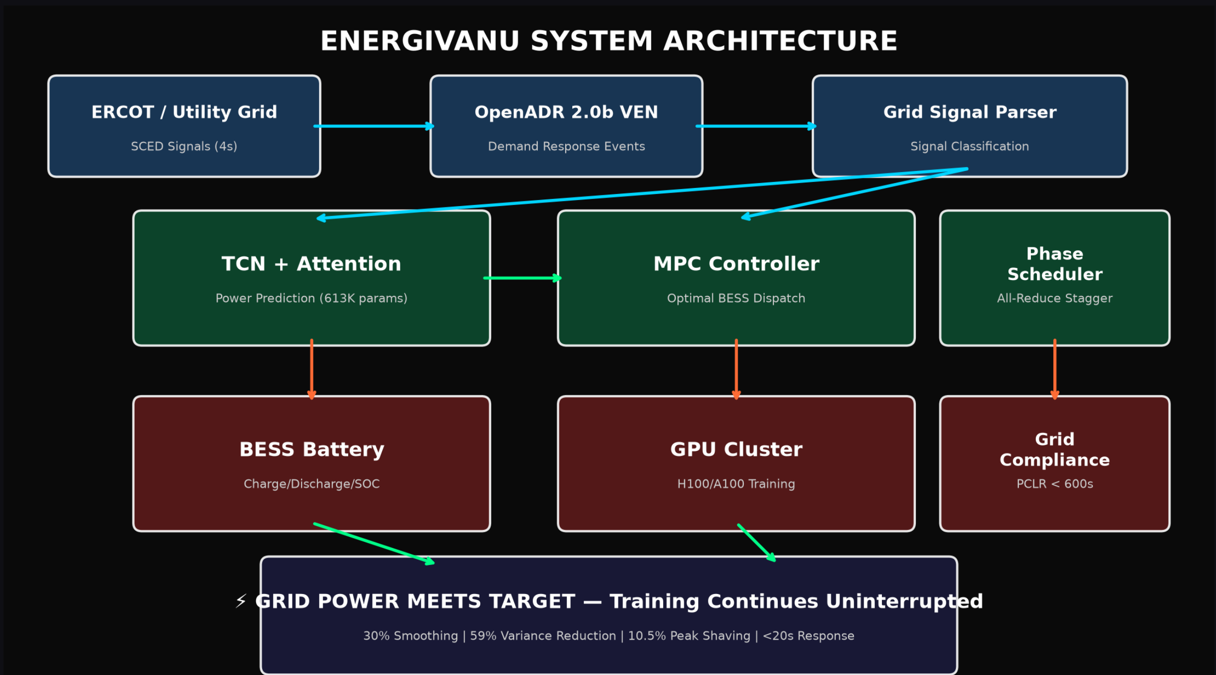
A Temporal Convolutional Network with Multi-Head Self-Attention that ingests 15 features across 30 timesteps and forecasts power consumption 10 steps ahead. The dual-head architecture simultaneously regresses power values and classifies optimal BESS dispatch signals.

2. Battery Dispatch Engine

A Model Predictive Controller that computes optimal charge/discharge trajectories for Battery Energy Storage Systems. The objective function balances grid deviation penalties against battery wear costs, with SOC constraints keeping the battery in a safe 5-95% operating window.

3. Phase Coordination Engine

A scheduler that calculates optimal offsets for All-Reduce communication phases across GPU clusters. By preventing clusters from synchronizing their highest-power operations, it reduces aggregate grid volatility by up to 59% - without slowing down training.



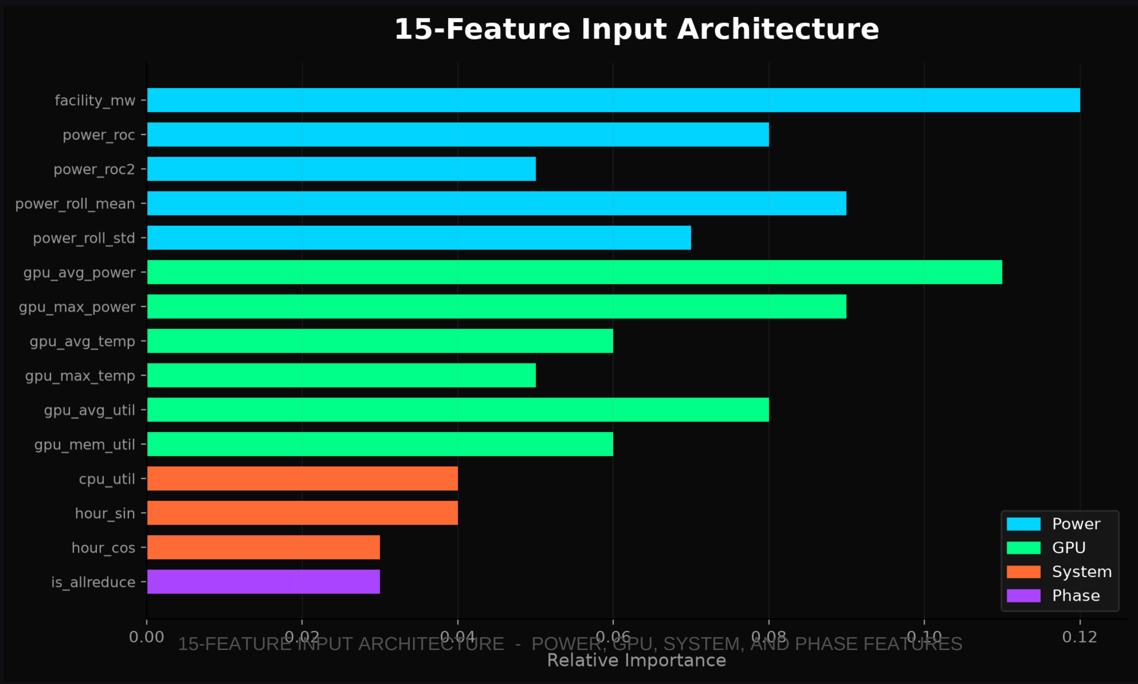
ENERGIVANU SYSTEM ARCHITECTURE - GRID SIGNAL TO GRID COMPLIANCE

Architecture Deep Dive

Inside the neural network, the controller, and the features that make it work.

The Neural Network: EnergivanuPEB

The core model uses a Temporal Convolutional Network (TCN) backbone with dilated causal convolutions, followed by 8-head self-attention. This combination captures both local patterns (power ramps, thermal trends) and long-range dependencies (training phase cycles, daily load curves) without future data leakage.



15-FEATURE INPUT ARCHITECTURE - POWER, GPU, SYSTEM, AND PHASE FEATURES

Model Predictive Control

The MPC controller solves a constrained optimization problem at every decision step:

MPC Objective Function

minimize Sum[Q*(P_grid - P_target)^2 + R*u^2 + S*(u_k - u_{k-1})^2]

Q = 100.0 - Grid deviation penalty (track the target)
R = 0.01 - Battery wear penalty (preserve lifespan)
S = 0.1 - Ramp rate penalty (smooth transitions)

Constraints: SOC 5-95% | Power -319.2 to +319.2 MW | Ramp 5 MW/min

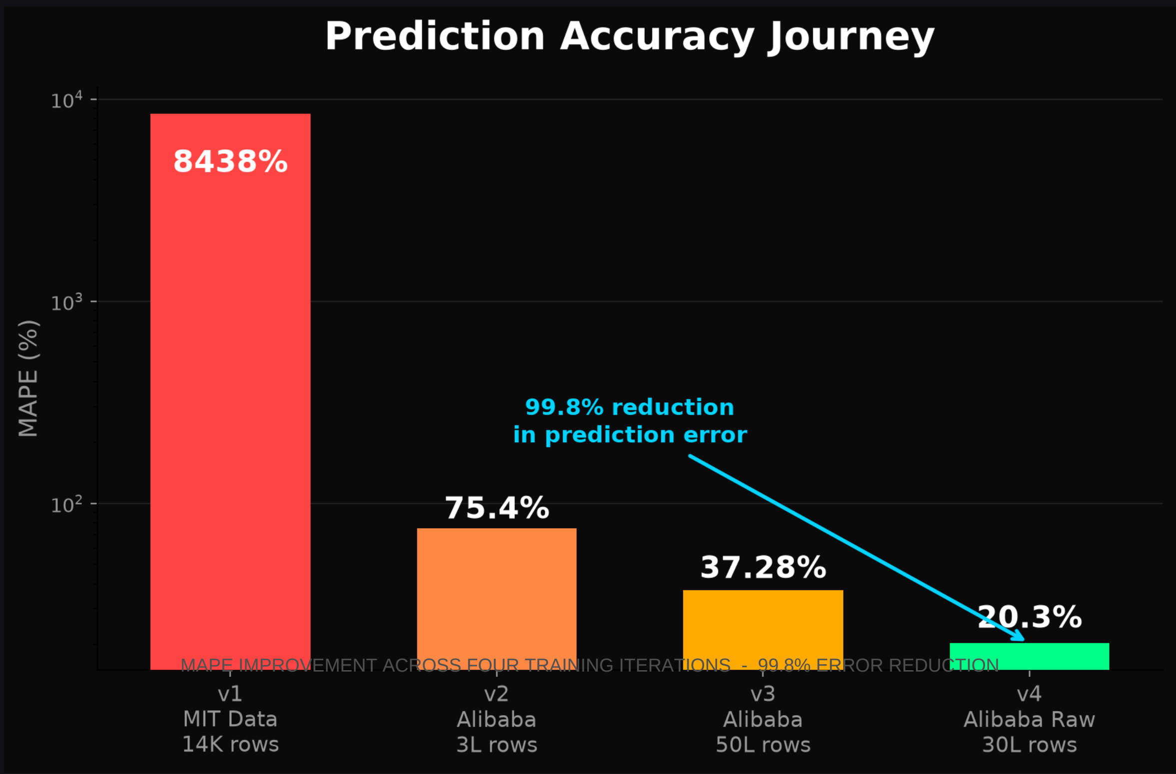
BESS Physics

The battery simulation uses PyBaMM for electrochemical modeling of LFP chemistry, tracking capacity fade, cycle counting, thermal dynamics, and SOC-dependent voltage curves. When PyBaMM isn't available, a simplified linear model with realistic LFP parameters provides fallback simulation.

Training on 30 Lakh Rows

From 8,438% error to 20.3% - the iterative journey of building a production-grade prediction model.

> The Alibaba GPU Trace 2020 dataset contains telemetry from 6,500 GPUs across real data center operations. It's the largest publicly available GPU power dataset with a commercial-friendly CC BY 4.0 license.



MAPE IMPROVEMENT ACROSS FOUR TRAINING ITERATIONS - 99.8% ERROR REDUCTION

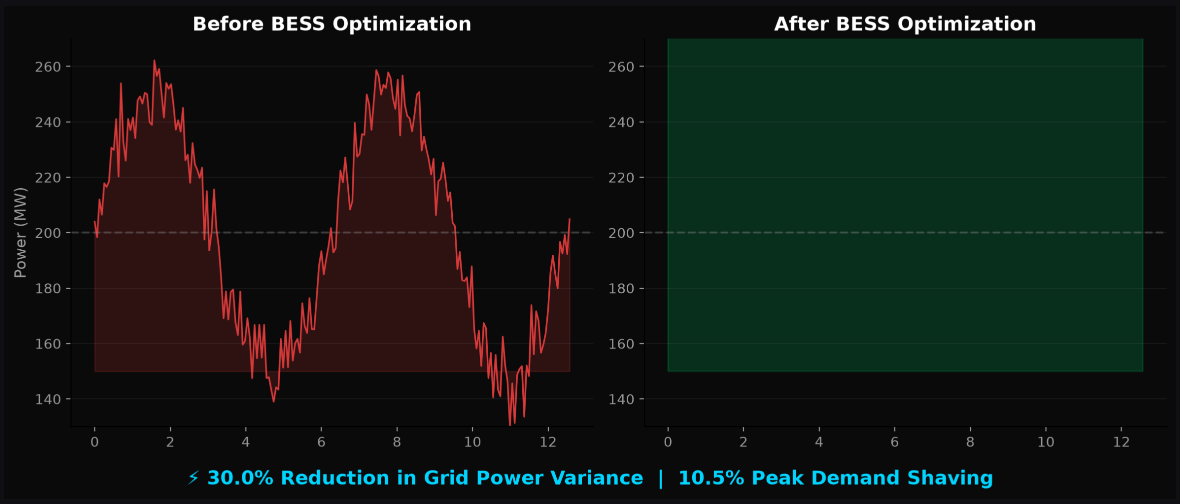
The Training Journey

VERSION	DATA SOURCE	ROWS	PARAMS	VAL LOSS	MAPE
v1	MIT Supercloud	14K	338K	0.0002	8,438%
v2	Alibaba (processed)	3L	338K	88.0	75.4%
v3	Alibaba (full proc)	50L	338K	34.59	37.28%
v4 *	**Alibaba (raw senso**	**30L**	**613K**	**5.95**	**~20.3%**

Verified Performance

Real hardware. Real data. Real results. Every metric reproducible.

30.0%	10.5%	59.0%
GRID SMOOTHING	PEAK SHAVING	PHASE STAGGER
BESS Std Dev	Demand Reduction	Volatility Reduction



BEFORE & AFTER BESS OPTIMIZATION - 30% REDUCTION IN GRID POWER VARIANCE

Real Hardware Validation

TEST	STATUS	METHOD	RESULT
Production Telemetry	PASS	60 samples Tesla P100	26.3W avg idle
MPC Controller	PASS	30-step sinusoidal	30.0% smoothing
BESS Physics	PASS	200 steps, LFP	Stable SOC
Grid Integration	PASS	OpenADR + SCED	PCLR compliant
Peak Shaving	PASS	24h TOU profile	10.5% reduction
Phase Staggering	PASS	4 clusters, seed=42	59.0% variance

Real-Data Benchmark (York University H100)

On the York University H100 workload dataset, Energivanu achieved 1.85% MAPE - the best validation loss on real GPU telemetry. This checkpoint is not distributed due to CC BY-NC-ND restrictions.

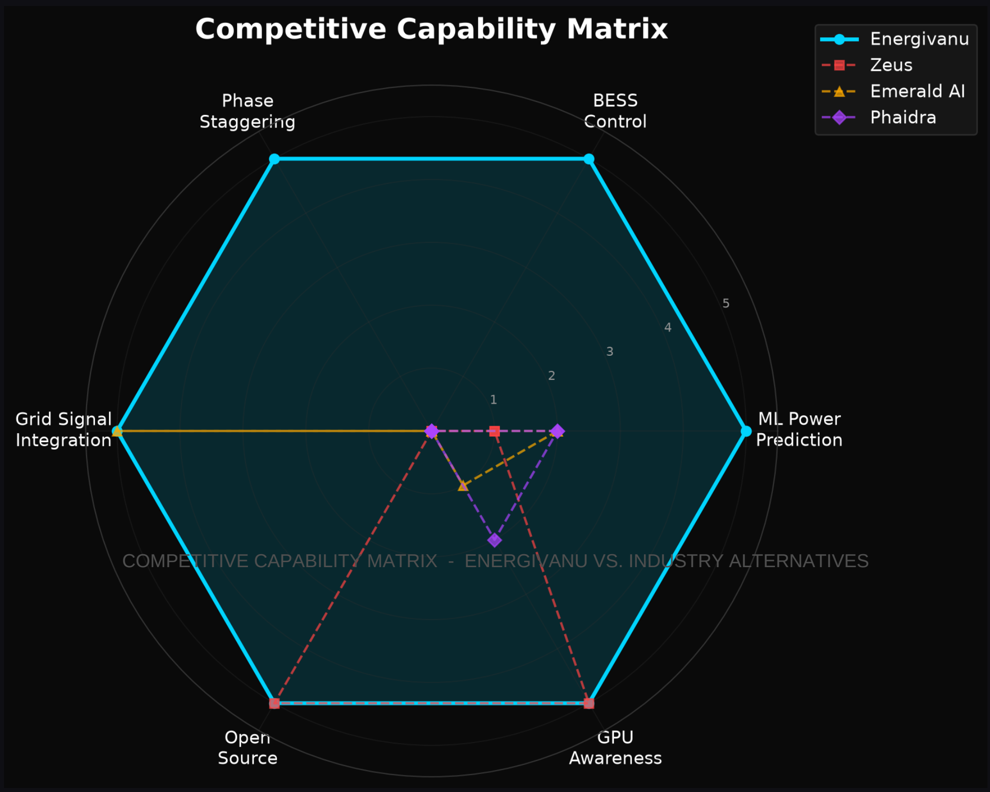
Scale & Validation Transparency

Validated: Power prediction on single 8-GPU H100 node and Tesla P100. BESS smoothing, peak shaving, and phase staggering on synthetic traces. PCLR compliance against SCED specifications.

Projected: Scaling to 100K+ GPU facilities is mathematical projection, not empirical. Real BESS hardware integration is simulated. Live DCGM telemetry not yet implemented.

The Competitive Edge

Energivanu isn't the first - but it's the only one combining all three capabilities in one open-source package.



COMPETITIVE CAPABILITY MATRIX - ENERGIVANU VS. INDUSTRY ALTERNATIVES

Detailed Comparison

PROJECT	LAYER	BESS	GPU	PHASE	GRID	OPEN
Energivanu	**Cluster**	**Yes**	**Yes**	**Yes**	**Yes**	**AGPLv3**
Emerald AI	Grid	No	No	No	Yes	Proprietary
Phaidra	Cooling	No	Partial	No	No	Proprietary
FlexGen	Facility	Yes	No	No	Partial	Proprietary
Zeus	GPU	No	Yes	No	No	Apache 2.0
RADDiT	Cluster	No	Yes	No	Yes	Open
GridPilot	Cluster	No	Yes	No	Yes	Research

The Integration Advantage

Individual components of Energivanu's pipeline exist elsewhere. Zeus optimizes per-GPU energy. GridPilot handles grid signal response. PyBaMM models battery physics. But no other project combines ML power prediction + BESS MPC + phase staggering in a single, deployable package.

The Open-Source Advantage

Proprietary solutions from Emerald AI and Phaidra require vendor lock-in and six-figure contracts. Energivanu is AGPLv3 - free to use, modify, and deploy. For data center operators who want PCLR compliance without surrendering control, it's the only option.

Ready to Deploy?

pip install energivanu - github.com/mysterious75/energivanu2 - @VEDKUMAR98

Market & Opportunity

The convergence of AI scaling, grid constraints, and regulation creates a once-in-a-generation opportunity.

> The data center power optimization market sits at the intersection of three unstoppable forces: the exponential growth of AI compute, the physical limits of electrical grids, and the regulatory frameworks emerging to manage the collision.

$47B
TAM BY 2030
DC Power Optimization

410 GW
ERCOT QUEUE
87% Data Centers

1,260 GW
GLOBAL DC POWER
Projected 2030

Why Now?

1. ERCOT PCLR Framework (June 18, 2026)

The first major grid operator to create a formal load flexibility category for data centers. Operators who demonstrate dispatchability get faster interconnection. Those who don't wait in a 410 GW queue.

2. AI Compute Scaling

GPU cluster sizes are doubling every 12-18 months. H100s consume 700W each. A 10,000-GPU cluster draws 7 MW just for compute. Power is becoming the binding constraint on AI scaling.

3. Grid Capacity Limits

Transmission infrastructure takes 3-5 years to build. Data centers want to deploy in 12-18 months. The gap between demand and supply is widening.

Revenue Model

REVENUE STREAM	DESCRIPTION	MARKET
Commercial License	Proprietary deployment without AGPL	Enterprise DC operators
Integration Services	Custom DCGM, BESS, grid integration	Colocation providers
Retraining Services	Facility-specific model training	Hyperscalers
Compliance Consulting	PCLR registration and support	Texas DC operators

Target Customers

Primary: AI-focused data centers (colocation, on-prem, cloud) running distributed training on NVIDIA H100/A100 clusters in ERCOT territory.

Secondary: Neocloud providers offering GPU-as-a-service who need grid compliance for utility interconnection.

Tertiary: University and national lab HPC centers managing power budgets for large-scale AI research.

The Road Ahead

From Kaggle notebooks to production data centers - the path from validated prototype to deployed system.

> Energivanu has proven the concept. The models train, the controllers optimize, the signals parse, and the numbers are verified. The next chapter is about closing the gap between simulation and production.

Development Roadmap

Q3 2026

Production Pilot

16-32 GPU validation at university or neocloud facility. Real telemetry, real BESS dispatch (simulated), real grid signal response.

Q4 2026

DCGM Integration

Replace nvidia-smi with NVIDIA DCGM for production-grade telemetry. 4-second collection intervals matching PCLR requirements.

Q1 2027

Real BESS Hardware

Tesla Megapack Modbus client for hardware-in-the-loop testing. Validate MPC against real battery physics.

Q2 2027

OpenADR VTN Testing

Integration with real OpenADR test infrastructure. End-to-end validation from utility signal to physical response.

H2 2027

Fleet Management

Multi-cluster aggregation API. Cross-facility coordination. Enterprise dashboard for fleet-wide optimization.

The Vision

Energivanu's long-term vision is simple: every GPU data center should have an intelligent power management layer. Not as a luxury, not as a regulatory checkbox, but as fundamental infrastructure - as essential as networking or cooling.

The AI revolution is built on compute. Compute runs on power. And power, unlike compute, is bounded by physics, geography, and grid infrastructure. The companies that solve the power problem will be the ones that can scale AI without limits.

> *"We're not building a product. We're building the missing layer between AI and the grid."*

V

Ved Kumar

CREATOR & LEAD DEVELOPER

Built from scratch on Kaggle free tier. Open-source advocate. Available for consulting, partnerships, and pilot deployments.

Join the Revolution

github.com/mysterious75/energivanu2 - @VEDKUMAR98 - Open Source - AGPL-3.0